

Bamboozling Certificate Authorities with BGP

Bamboozling Certificate Authorities with BGP - Henry Birge-Lee, Yixin Sun, Anne Edmundson, Jennifer Rexford, Prateek Mittal, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity18/presentation/birge-lee>>
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Bamboozling Certificate Authorities with BGP

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Abstract:

The Public Key Infrastructure (PKI) protects users from malicious man-in-the-middle attacks by having trusted Certificate Authorities (CAs) vouch for the domain names of servers on the Internet through digitally signed certificates. Ironically, the mechanism CAs use to issue certificates is itself vulnerable to man-in-the-middle attacks by network-level adversaries. Autonomous Systems (ASes) can exploit vulnerabilities in the Border Gateway Protocol (BGP) to hijack traffic destined to a victim's domain. In this paper, we rigorously analyze attacks that an adversary can use to obtain a bogus certificate. We perform the first real-world demonstration of BGP attacks to obtain bogus certificates from top CAs in an ethical manner. To assess the vulnerability of the PKI, we collect a dataset of 1.8 million certificates and find that an adversary would be capable of gaining a bogus certificate for the vast majority of domains. Finally, we propose and evaluate two countermeasures to secure the PKI: 1) CAs verifying domains from multiple vantage points to make it harder to launch a successful attack, and 2) a BGP monitoring system for CAs to detect suspicious BGP routes and delay certificate issuance to give network operators time to react to BGP attacks.

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@inproceedings {217541, author = {Henry Birge-Lee and Yixin Sun and Anne Edmundson and Jennifer Rexford and Prateek Mittal}, title = {Bamboozling Certificate Authorities with {BGP}}, booktitle = {27th {USENIX} Security Symposium ({USENIX} Security 18)}, year = {2018}, isbn = {978-1-931971-46-1}, address = {Baltimore, MD}, pages = {833--849}, url = {https://www.usenix.org/conference/usenixsecurity18/presentation/birge-lee}, publisher = {{USENIX} Association}, }
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